$$\frac{n-r}{n} = 1 - \frac{r}{n}$$

$$\because e^{-x} \approx 1 - x$$

$$\therefore \frac{n-r}{n} \approx e^{-\frac{r}{n}}$$

No shared birthdays among k guests:

$$\begin{aligned} P_0 &= 1 \cdot \frac{364}{365} \cdot \frac{363}{365} \cdots \frac{365-k+1}{365} \\ &= \prod_{r=1}^{k-1} \frac{365-r}{365} \\ &\approx \prod_{r=1}^{k-1} e^{-\frac{r}{365}} \\ &= e^{-\frac{1}{365}\sum_{r=1}^{k-1} r} \\ &= e^{-\frac{k(k-1)}{730}} \end{aligned}$$

$$P(\text{at least two share a birthday}) \approx 1 - e^{-\frac{k(k-1)}{730}}$$

$$1 - e^{-\frac{k(k-1)}{730}} \geq \frac{1}{2}$$

$$\frac{k(k-1)}{730} \geq \ln 2$$

$$k(k-1) \geq 730 \cdot \frac{253}{365} = 506$$

$$k^2 - k - 506 \geq 0 \implies (k-23)(k+22) \geq 0$$

$$k \geq 23$$

Let h be the number of guests
No guest has the same birthday as the host:

$$\left(\frac{364}{365}\right)^h = \left(\frac{365-1}{365}\right)^h \approx e^{-\frac{h}{365}}$$

$$P(\text{at least one guest matches the host}) \approx 1 - e^{-\frac{h}{365}}$$

$$1 - e^{-\frac{h}{365}} \geq \frac{1}{2} \implies \frac{h}{365} \geq \ln 2$$

$$h \geq 365 \cdot \frac{253}{365} = 253$$

This solution was submitted by **Jon**

Published and refined by **Jon**

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